Customer Pattern Discovery

Segmentation, dimensionality reduction, and association-rule mining

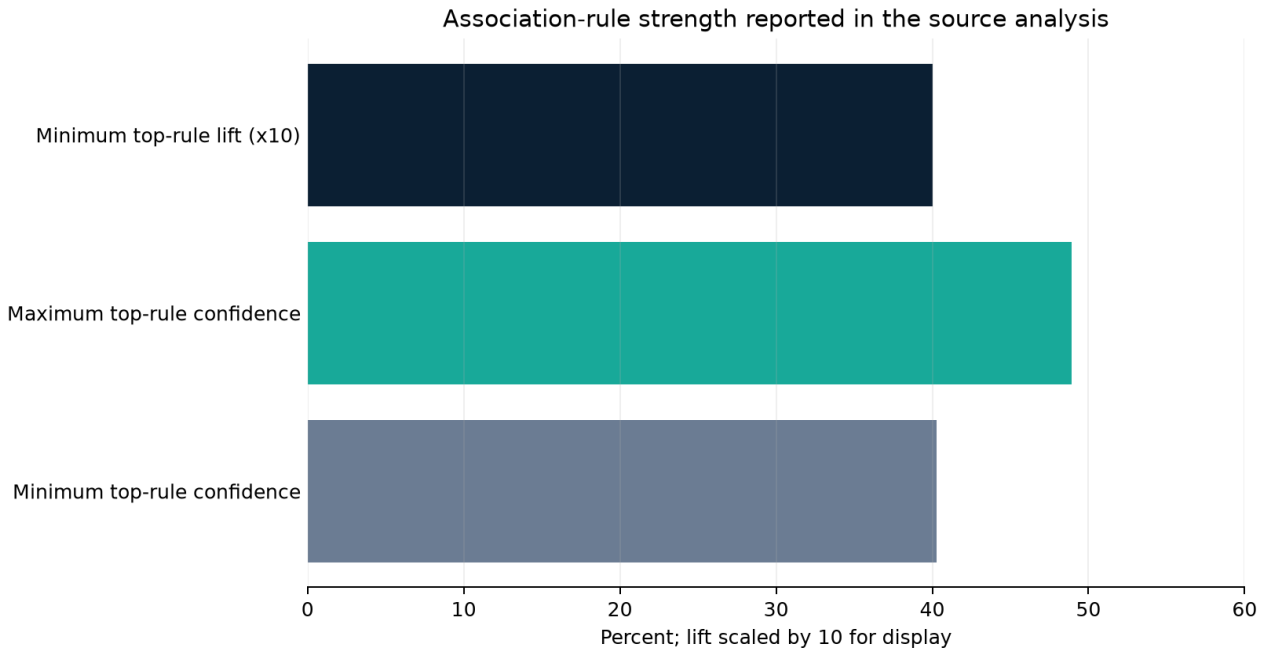
Portfolio Case Study

Executive Summary

This portfolio case study consolidates three complementary analyses into one discovery workflow: K-means segmentation, principal component analysis, and Apriori association rules. The source analyses used cluster diagnostics, retained six principal components under the Kaiser criterion, and surfaced product rules with confidence from 40.3% to 48.9% and lift above four. Together, the methods convert high-dimensional behavior into interpretable segments and testable merchandising hypotheses.

Decision Question

How can unsupervised learning reveal customer segments, simplify behavioral features, and identify cross-sell relationships?



Methods

- Standardized continuous customer features before distance-based segmentation.
- Used elbow and silhouette diagnostics to evaluate K-means solutions.
- Applied PCA to reduce correlated behavioral variables and retained six components under the Kaiser criterion.
- Encoded transactions and mined frequent itemsets and association rules using support, confidence, and lift.

Evidence and Findings

Evidence	Observed result
Segmentation	K-means produced interpretable profiles for comparing behavioral patterns and churn exposure.
Dimension reduction	Six components were retained under the eigenvalue-greater-than-one criterion.
Association confidence	Top source-analysis rules ranged from 40.3% to 48.9% confidence.
Association lift	Top relationships exceeded lift of four, with the strongest approaching six.

Limitations

- K-means favors roughly spherical, similarly scaled clusters and is sensitive to initialization.
- PCA components trade original-variable simplicity for compact representation.
- Association rules describe co-occurrence, not causality or incremental campaign lift.
- The public repository excludes the restricted source dataset and provides reusable methods instead.

Recommended Next Actions

- Profile clusters with stable business measures before assigning operational labels.
- Treat association rules as hypotheses for controlled cross-sell tests.
- Re-evaluate cluster and rule stability on refreshed data before activation.

Reproducibility

The companion repository contains typed Python modules, tests, dependency declarations, data-access notes, and a rendered analysis notebook. Restricted source data and the working notebook are intentionally excluded from the public repository.

Results are reported as historical analytical evidence. They should be reproduced on an authorized equivalent dataset before operational use.